DAMON Updates and Plans:

Tiering, Monitoring, DAMON-X, and THP

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42 authors, 469 commits since 2025-03-26 (LSFMMBPF 2025)

DAMON in Nutshell

DAMON in One Sentence

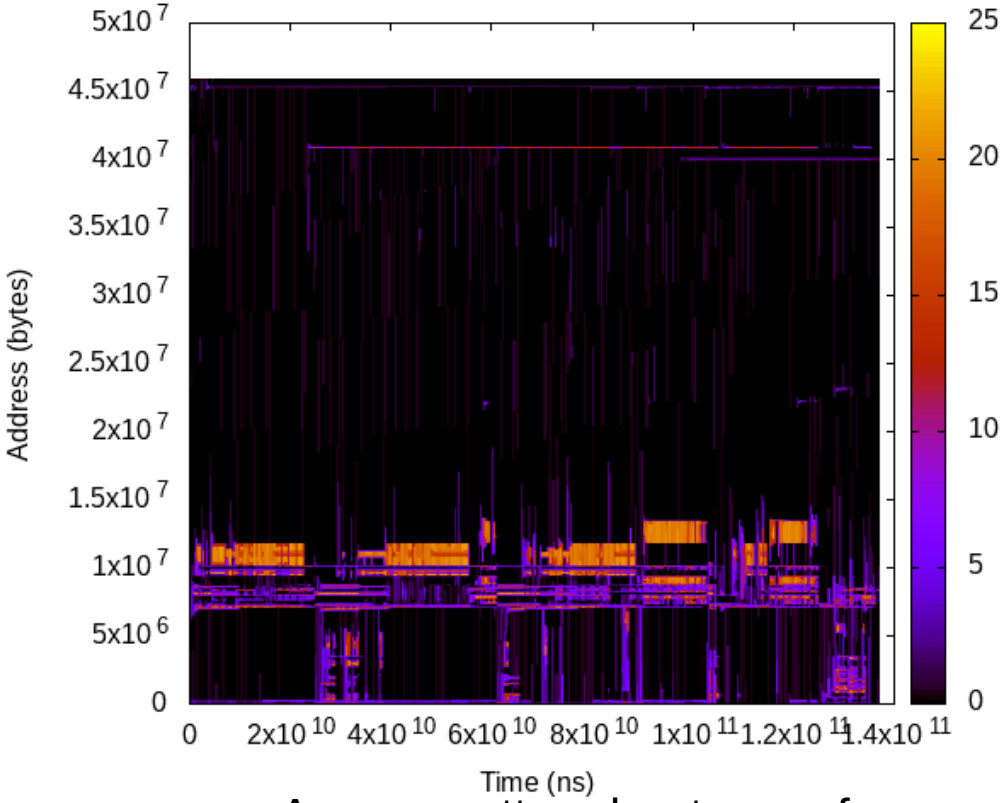
“DAMON is a Linux kernel subsystem for efficient access monitoring and access-aware system operations.”

High Level Implementation

- Spawn a kernel thread that does
- See if each page is accessed, every 5ms
- Inform users the findings, every 100ms
- Time intervals can be [auto-]tuned

Providing Access Information

- Location
- Frequency
- Stability
- Recency



Access pattern heatmap of Splash2x/Raytrace

Design

- Goals: Accurate, Lightweight, Tunable and Automated
- Mechanisms: Sampling, Adaptive regions, Sysfs interface, Feedback loops
- Results
  - <0.1% CPU overhead on products

Availability and Usages

- Available on >=5.15 upstream kernels
- Enabled on most distro kernels
- Being used for products and researches

DAMOS: Second Face of DAMON

- DAMon-based Operation Schemes
  - “Page out cold memory”
  - “Migrate hot/cold pages to DRAM/CXL node”
- Turn DAMON into access-aware memory management system operations engine

Community

- Public channels
  - Mailing list: damon@lists.linux.dev
  - Project website: https://damonitor.github.io/
- Private channels
  - Maintainer email: sj@kernel.org
  - DAMON Beer/Coffee/Tea Meetup

DAMON in One Sentence

“DAMON is a Linux kernel subsystem for efficient access monitoring and access-aware system operations.”

Table Of Contents (55 mins)

- Updates from LSFMMBPF 25
  - Memory Tiering (10 mins)
  - Data Attributes Monitoring (10 mins)
- New for LSFMMBPF 26
  - Page Faults or Alternatives for DAMON (15 mins)
  - DAMON-X (5 mins)
  - Access-aware THP (15 mins)

Memory Tiering

LSFMMBPF’25 Recap

- SK hynix developed DAMOS_MIGRATE (6.11)
  - For CXL capacity extension of HMSDK
  - DAMOS_MIGRATE hot/cold data DRAM ↔ CXL
  - Static hot/cold threshold
- SJ was developing TPP-DAMON
  - Auto-tune thresholds for high DRAM utilization
  - Achieved ~4.42% Taobench score increase

Multi-thread Tiering (SK hynix)

- Per-subregion DAMON worker thread
  - Accelerate hot/cold pages finding
  - Accelerate migrations
- Achieves ~94% llama.cpp speedup

Dynamic Interleaving (Micron)

- Extend DAMOS_MIGRATE
  - Multiple destination nodes with their weights
  - For dynamic interleave weights
  - Works on virtual address only
- Achieves ~25% benchmark speedup
- Merged into 6.17

Interleaving Auto-tune (Micron)

- Add new DAMOS auto-tune goal
  - For physical address based dynamic interleaving
  - Example usage: x:y hot memory interleaving
    - Migrate hot data CXL→DRAM for X% hot data in DRAM
    - Migrate hot data DRAM→CXL for Y% hot data in CXL
- Merged into mm-new after 7.1-rc1

TPP-DAMON

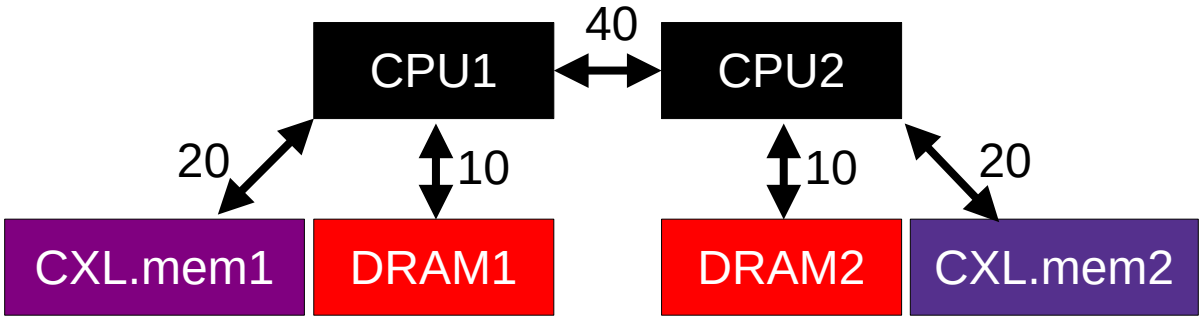
- Merged into 6.16
- Cgroup-aware extension developed (6.19)
- Positive test results privately shared
- No more active development is planned (Support mode)

NUMA-TPP-DAMON

- Tiering: special case NUMA management
  - Single accessor (CPU node)
  - Single promotion/demotion path
- General NUMA: N Accessors
  - CPU, GPU, *PU
  - N promotion/demotion paths
  - Idea: run accessor-aware TPP for each path

NUMA-TPP-DAMON in Example

- TPP1 migrate CPU1-hot/cold memory
  - for DRAM1 ↔ CXL.mem1 ↔ DRAM2 ↔ CXL.mem2 path
- TPP2 migrate CPU2-hot/cold memory
  - for DRAM2 ↔ CXL.mem2 ↔ DRAM1 ↔ CXL.mem1 path



Possible NUMA topology

	CPU1	CPU2
DRAM1	10	50
CXL.mem1	20	60
DRAM2	50	10
CXL.mem2	60	20

Relative access speed from CPU to memory

NUMA-TPP-DAMON Status

- Brainstorming level early planning
  - Idea is shared on LPC’25 special-mm uconf
- Challenge: per-accessor monitoring

Inputs?

- ~9:45

Data Attributes Monitoring (was: light-weight cgroups monitoring)

LSFMMBPF’25 Recap

- Meta developed page level monitoring
  - Show how many bytes in each DAMON-region
    - Hugepage, belong to specific cgroup, etc
- Challenge: High Overhead
  - Filters are applied in folio level

Page Level Monitoring Updates

- Essential changes are merged into 6.15
- Meta extended hugepage support for virtual address spaces (merged into 6.18)
- Idea for fleet-wide monitoring is shared
  - Machine-level sampling

Data Attrs Monitoring

- Sampling-based page level monitoring
- Let users register multiple probes
  - Each probe is for each data attribute to monitor
  - Each probe constitutes DAMOS filter-like filters
- For each access-sample page, apply probes
- Count/expose per-region per-probe hits

Attrs Monitoring: Outcomes

- Lightweight, Scalable
  - Access sampling is reused
  - Region-adjustment is applied
- Accuracy is arguable
  - Usefulness depends on multiple factors
  - Page level monitoring could be used together

Attrs Monitoring: Status

- First RFC is shared
  - Supports page anonymity monitoring
- Aiming to drop RFC with
  - More stabilized user interface
  - Cgroup support

Attrs Monitoring: Future Plan

- Data access as a data attribute
  - Add pg_idle DAMON filter
  - Allow attr-based region merge/split
- Extend for richer access check primitives
  - Add DAMON filters for page faults, PMU, etc
  - Path forward to NUMA-TPP-DAMON

Inputs?

- ~9:55

Page Faults or Alternatives for DAMON

Motivation

- Per-NUMA/CPUs/threads monitoring
  - For cache-aware sched_ext
  - For NUMA-TPP-DAMON
- Per-reads/writes monitoring
  - For easiest live migration target VM finding
  - For easiest KSM merge candidates finding

Challenge: Lack of Information

- pg_idle: DAMON’s access check primitive
  - Lacks accessor, type of access information
- Solution: use other primitives
  - Page fault events
  - PMU (e.g., AMD IBS)

NUMA Hinting Faults

- NUMA balancing’s access check primitive
- Mark a page not exists (PTE) but accessible (vma)
- Page fault handler notifies faults on marked pages to NUMA balancing

NUMA Hinting for DAMON

- Prototype with (very dirty) hack exists
- Problems
  - Making page fault handling code dirty
  - Interference between DAMON and NUMA balancing

NUMA Hinting Cleanup

- Cleanup should come first
- Apparently no doubt todo
- Uffd workingset detection does a sort of cleanup

NUMA Hinting Interference

- NUMA balancing and DAMON can show faults that are caused by each other
- Intuitively odd behavior

Option 1: Build-time Isolation

- Disallow building NUMA balancing and page fault-mode DAMON together
- Advantage: simple
- Disadvantage: inflexible

Option 2: Runtime Isolation

- Let them exclusively enabled in runtime
- Ask them to uninstall installed marks before disabled

Option 3: Partial isolation

- Let them exclusively enabled in runtime
- Let them show each other-installed protections
  - In a sense, NUMA balancing interferes past/future enabled instances already

Option 4: Let Them Interfere

- Inform faults to both NUMA balancing and DAMON if they are enabled
- Any real problem from the interference?
  - No problem from DAMON’s perspective
- Concurrent usages would be rare anyway

Proposal

- Cleanup is must-do
- Start with option 1 (build time isolation)
  - Or option 4 (let them interfere)
- And revisit if users request changes

Alternative: Perf Events

- Multiple RFC implementations available
- perf maintainers show no concern (yet)
- Problem: Work on only specific h/w

Inputs?

- ~10:10

DAMON-X

Why: DAMON That Just Works

- DAMON goal: Tunable and Automated
- DAMON sysfs: For flexible user controls
- DAMON modules: For users who want it just works
  - E.g., DAMON_RECLAIM and DAMON_STAT
  - Should just work with 1-2 parameter setup

Problems in DAMON Modules

- Runs in exclusive manner
  - Cannot run >=2 modules at same time
- Inflexible
  - Flexible debugging/testing purpose interface is needed on production

Ideas

- Modules share monitoring params
  - The recommended intervals auto-tuning setup
  - Runs for the physical address space
- Differences are only DAMOS schemes
- Single DAMON context can run multiple schemes
- Online schemes [un]install is always supported

DAMON-X

- DAMON-Enabled Compute System
- Always run common setup DAMON
- Modules online-[un]install DAMOS schemes
- Support optional sysfs interface

Project Status

- Idea was shared on LPC’24 MM microconf
- Planning to POC-implement by Q3’26

Inputs?

- ~10:15

Access-aware THP

Motivation

- THP is known to be
  - Good: makes it faster (TLB miss reduction)
  - Bad: makes it larger (internal fragmentation)
- madvise() allows users wisely use THP
- Could DAMOS make it access-aware?

Initial Implementation and Test

- DAMOS_[NO]HUGEPAGE hot/cold pages
  - Hot: DAMON found any access
  - Cold: Region-sz >=2MiB, Unaccessed >=7 secs
- Test Results with PARSEC 3.0
  - Removes 80% memory bloat from THP
  - Preserves 46% THP performance gain
  - Numbers are not very stable

Productionizing Challenges

- Was unable to find good answers to
- Collapse? Split? Both?
- Virtual or physical address space?
- What is the ideal hot/cold thresholds?

Huawei’s Recent Approach

- Find three CPU hogger processes
  - Periodically change the list of processes
- Collapse hot memory in the processes
  - Auto-tune thresholds until make progress
- Good results on MySQL
- Still no good answers to the questions

Collapse or Split

- Should we collapse?
  - Or, enable THP=always?
- Should we split?
  - Or, rely on THP shrinkers?
    - Or, help or override THP shrinkers?
- Should we do both?
  - Or, rely on other mechanisms and do only one?

Virtual or Physical Address?

- Virtual addr is essential for collapsing
  - What processes to pick? Let users decide?
  - How to deal with DAMON-X (physical addr)?
- Physical addr can be used for splitting
  - Do not interfere with DAMON-X
  - Do not require users control

How to Set Thresholds?

- Quota goal-based autotune? What should be the goal?
  - huge:non-huge page ratio?
    - Huawei recently proposed this
  - TLB miss rate?
  - Memory bloat or pressure?

Thoughts At The Moment

- Collapsing for wise user makes sense
  - Huawei mm.git-merged DAMON_COLLAPSE
- Splitting for DAMON-X makes sense
- New DAMOS autotune goal makes sense for both

Better-help madvise()?

- DAMON is difficult to use
  - Require root permission
  - DAMON-X is for physical address space
- DAMON-X may better help

DAMON-X for easier madvise()

- User: request hotness of x-y address range of proc A
- DAMON-X: Sample-aggregate-output
  - pick N pages in the virtual address range
  - Find their physical addresses
  - Find hotness of the pages on its physical address-based snapshot
  - Aggregate their hotness and return to user

More Inputs?

- NUMA-TPP-DAMON
- Light-weight data attributes monitoring
- DAMON: Data Attributes Monitoring/Operations eNgine
- DAMON-X
- DAMON-THP

Backup Slides

Changes since LSFMMBPF’25

- 6.15-rc1 (2025-04-06)
  - DAMOS filters for hugepages, active, unmapped
  - DAMON intervals auto-tuning
- 6.16-rc1 (2025-07-08)
  - Self-tuned memory tiering
- 6.17-rc1 (2025-08-10)
  - DAMOS interleaving
  - DAMON_STAT
- 6.18-rc1 (2025-10-12)
  - 32bit PAE support
- 6.19-rc1 (2025-12-14)
  - Memcg-aware memory tiering
- 7.0-rc1 (2026-02-22)
  - DAMOS max_nr_snapshots
  - Active:inactive pages ratio based LRU sorting
- 7.1-rc1 (2026-04-26)
  - Quota goal tuner

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_free_bp 0.5% 0 \
        --damos_filter reject young \
    --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_hot 0 --damos_access_rate 5% max \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
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        --damos_filter reject young \
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        --damos_action migrate_hot 0 --damos_access_rate 5% max \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Start DAMON in kernel.

Dear DAMON user-space tool,

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s                                  200MB \
        --damos_filter allow node_mem
        --damos_filter allow young
    --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_hot 0 --damos_access_rate 5% max \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Create one DAMON worker thread, monitoring for node 0 (upper tier)

With auto-tuned monitoring intervals.

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_q[illegible] interval 1s --damos_quota_space 200MB \
        -[illegible] \
        -[illegible]
    --num[illegible] 5ms 10s \
        -[illegible]s_rate 5% max \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

And ask it to *migrate* un-accessed (access_rate *0%*) memory to node *1* (lower tier), *cold*est one first.
IOW, **demote** cold memory.

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_free_bp 0% 0 \
        --damos_filter reject young \
    --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_hot 0 --damos_access_rate 5% max \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

And do that once per second.

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_free_bp 0.5% 0 \
        --damos_fi[illegible]ject young \
    --numa[illegible]
        --[illegible]x \
        --[illegible]
        --[illegible]
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Within *one second*, do the migration only up to *200 MiB* of memory.

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_free_bp 0.5% 0 \
        --damos_filter reject young \
    --numa_node 1 [illegible]ring_intervals_goal 4% 3 5ms 10s \
        --[illegible]x \
        --[illegible]
        --[illegible]
        --damos_quota_goal node_mem_used_bp 99.7% 0 \
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Under the 200MiB/1s hard limitation, auto-tune the real soft limitation to make node 0's free memory ratio becomes 0.5% soon.

TPP-DAMON in More Details

```
damo start \
    --numa_node 0 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_cold 1 --damos_access_rate 0% 0% \
        --damos_apply_interval 1s \
        --damos_quota_interval 1s --damos_quota_space 200MB \
        --damos_quota_goal node_mem_free_bp 0.5% 0 \
        --damos_filter reject young \
    --numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
        --damos_action migrate_hot 0 --damos_access_rate 5% max \
        --
        --
        --
        --damos_filter allow young \
        --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
    --nr_targets 1 1 --nr_schemes 1 1 --nr_ctxs 1 1
```

Btw, if the page is PG_Young, don’t do the migration.
DAMON might identify young page as not accessed.

TP

Create another DAMON worker thread that monitors *node 1* (lower tier) with auto-tuned monitoring intervals.
Ask it to migrate memory shown >=*5%* access rate to *node 0* (upper tier), *hot*test first. IOW, **promote** hot pages.
Do the migration once per second, up to 200 MiB of memory.
Under the *200 MiB/s* hard limit, adjust the real soft limit aiming *node 0* memory utilization ratio *99.7%*.
Btw, to that only if the page is PG_*Young*.

```
damo
-                                                                    % \
    --damos_                    node_mem_free_bp 0.5% 0 \
    --damos_filter reject young \
--numa_node 1 --monitoring_intervals_goal 4% 3 5ms 10s \
    --damos_action migrate_hot 0 --damos_access_rate 5% max \
    --damos_apply_interval 1s \
    --damos_quota_interval 1s --damos_quota_space 200MB \
    --damos_quota_goal node_mem_used_bp 99.7% 0 \
    --damos_filter allow young \
    --damos_nr_quota_goals 1 1 --damos_nr_filters 1 1 \
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```